

VINESH SHANMUGAM

ECE ENGINEER | AI/ML | EMBEDDED SYSTEMS | IoT | FULL-STACK DEVELOPMENT

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SUMMARY

Electronics & Communication Engineering student who builds intelligent, cloud-connected hardware-software systems spanning embedded devices, AI/ML, and full-stack web applications. Experienced in developing ESP32-based sensing pipelines, training and deploying machine learning models, and building REST APIs and React dashboards that turn real-world sensor data into actionable insight.

TECHNICAL SKILLS

Programming: Python, Java, C, C++, JavaScript, TypeScript, SQL

AI / Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Hugging Face, Computer Vision, TinyML

Embedded Systems / IoT: ESP32, ESP8266, Arduino, Sensors, I2C, UART, GPIO, MQTT

Frontend: React, TypeScript, Vite, Tailwind CSS, Chart.js, Recharts

Backend: FastAPI, Flask, Node.js, REST APIs, WebSockets

Database / Cloud: Firebase, PostgreSQL, SQLAlchemy

DevOps / Tools: Git, GitHub, Docker, CI/CD, Vercel, Render, VS Code, Arduino IDE, Google Colab

PROJECTS

TinyML-Based Human Activity & Fall Detection — ESP32 · Accelerometer · Python · Scikit-learn · TensorFlow · TinyML · Firebase

- Designed an end-to-end embedded AI pipeline that acquires accelerometer motion data on an ESP32, applies signal processing and feature extraction, and classifies human activity to detect falls in real time.
- Trained and evaluated a TinyML classification model using Scikit-learn and TensorFlow, then optimized it for on-device inference on resource-constrained embedded hardware.
- Integrated the ESP32 sensing node with Firebase to stream classification results to a cloud dashboard for real-time activity monitoring.

PancreaSense AI — React · TypeScript · FastAPI · PostgreSQL · SQLAlchemy · ESP32 · Machine Learning · SHAP

- Built a research-prototype healthcare AI platform combining ESP32-based sensor data acquisition with a FastAPI backend and PostgreSQL data layer for structured health-data management.
- Implemented a machine learning risk-assessment model with SHAP-based explainability to surface interpretable, feature-level insights for AI-assisted screening rather than diagnosis.
- Developed a React and TypeScript dashboard to visualize risk indicators and model explanations for end users of the experimental platform.

SmartWatts — ESP32 · Firebase · React · FastAPI · Machine Learning

- Engineered a cloud-connected IoT system for real-time electricity monitoring, using ESP32 nodes to collect consumption data and stream it to Firebase.
- Built a FastAPI backend and React dashboard for energy analytics, visualizing usage trends and flagging abnormal consumption patterns.
- Applied a machine learning component to generate intelligent, usage-based recommendations for more efficient energy consumption.

Transit Assist — Computer Vision · AI · Object Detection · Maps APIs

- Developed a computer vision-based transportation assistance system using object detection to identify transit vehicles and relevant surroundings in real time.
- Integrated navigation and mapping APIs to provide real-time route and transit information, with an accessibility-oriented interaction design.

DevOpsGuard — Python · FastAPI · Docker · GitHub · CI/CD

- Built a containerized DevOps automation tool with a FastAPI backend and REST API architecture to monitor development and deployment workflows, using Docker and GitHub-based CI/CD concepts.

EXPERIENCE

Wireless Communication Engineering Intern — BSNL

- Gained hands-on exposure to wireless communication infrastructure at a telecom service provider, including cellular network architecture, base station equipment, and signal transmission systems.
- Assisted in monitoring signal quality and network performance, and studied RF propagation and GSM/LTE network fundamentals in a live telecom environment under engineer supervision.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics & Communication Engineering

2024 – 2028

Meenakshi Sundararajan Engineering College, Chennai, Tamil Nadu

RESEARCH INTERESTS

TinyML · Edge AI · Intelligent IoT · Healthcare AI · Computer Vision · Embedded Intelligence · Signal Processing